

Predoc Empirical Assignment
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Abstract

This report replicates and extends the key findings of Card, Chetty, and Weber (2007) by using Austrian administrative data to assess how severance payments and extended unemployment benefits influence labor market transitions. The analysis employs a regression discontinuity (RD) design based on tenure thresholds, identifying causal effects on nonemployment durations, reemployment hazards, and subsequent match quality.

Chapter 1 presents Austrias strict regulations on dismissals and outlines the rules governing severance pay and unemployment benefits. It also explains the data cleaning process, sample construction, and core summary statistics, ending with evidence that baseline characteristics remain continuous at the relevant thresholds.

Chapter 2 replicates the original analysis by first showing graphical evidence on the distribution of layoffs around the three-year cutoff, worker histories, and wage patterns. It then examines nonemployment durations, adjusting for censoring, and highlights how severance pay and extended benefits alter reemployment patterns.

Chapter 3 provides additional robustness checks, investigating whether baseline covariates exhibit any unexpected jumps near the cutoff. It also discusses bandwidth selection and local polynomial methods used to confirm the RD designs validity.

Finally, Chapter 4 introduces a Cox proportional hazard model to handle censored unemployment spells and time-varying job search intensities. The results suggest both severance pay and extended benefits delay reemployment by providing extra liquidity but do not significantly improve post-unemployment wages or subsequent job durations.

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Chapter 1

Institutional Background and Identification Validity

This study replicates the main findings of Card, Chetty, and Weber (2007), examining how severance payments impact unemployment durations using Austrian labor market data. Austria's tightly regulated employment termination process provides an ideal setting for applying a regression discontinuity (RD) approach to evaluate causal effects.

1.1 Institutional Context

Employment termination regulations in Austria are stringent. Employers at firms with at least five employees must consult with a works council, youth council, or disabled workers' representatives before initiating any dismissal. After notification, the council is allowed a few working days to provide input, after which, if disagreement persists, the dismissal can be challenged in the Court of Employment and Social Affairs. Notice periods for termination depend on tenure; employees with less than two years of tenure require a six-week notice, while those exceeding 25 years require up to five months. Waged employees under industrial codes typically have a fixed notice period of 14 days.

In 1993, Austria introduced the Employment Contract Adjustment Law, significantly impacting large-scale layoffs by ensuring that employment contracts are upheld during company takeovers unless the company faces bankruptcy. Additionally, the Public Employment Service provides short-time working allowances during temporary company difficulties, requiring a minimum employment threshold of 20

Eligibility for severance pay requires a minimum tenure of three years, with amounts increasing proportionally to tenure. Employees voluntarily resigning may still qualify under specific conditions,

such as reaching pension eligibility or maternity leave conditions. Severance pay typically equates to two months' salary for employees with three years of service.

Unemployment insurance covers most workers, excluding civil servants and self-employed individuals. The unemployment benefit amount is determined by social security contributions over the preceding two years, typically equating to approximately 55% of net income. The benefit duration varies from 20 to 52 weeks, based on age and prior employment duration. Unemployment assistance is available post exhaustion of unemployment benefits and is means-tested, capped at 92% of prior unemployment benefits.

1.2 Data Cleaning and Sample Selection

The dataset mirrors the criteria established by Card et al. (2007), selecting individuals aged 20 to 50, excluding construction workers, voluntary job leavers, and those rehired by previous employers. Observations span unemployment periods ending between 1980 and 2001, with further restriction to individuals whose previous job tenure was between one and five years.

Table 1.1 summarizes the data adjustments:

Table 1.1: Data Cleaning and Sample Selection

Exclusion Criteria	**Affected Data (%)**
Age <20 or >50 at job termination	(Already Cleaned)
Construction workers	(Already Cleaned)
Voluntary quitters	8.0
Recalled to prior firm	18.0
Job separation outside 1981-2001	15.1
Employment duration <1 year	0.0
Employment duration >5 years	28.6

Note: Age restrictions were applied to avoid special programs for older workers. Construction workers were excluded due to separate severance regulations. Voluntary quitters were removed due to ineligibility for severance pay. Recalled workers were excluded as they have lower incentives to actively seek new employment. The sample was restricted to job separations within 1981-2001 to ensure temporal consistency. Employment duration constraints ensured eligibility for unemployment insurance and maintained homogeneity around the RD threshold.

Employment duration was categorized monthly around the severance pay threshold, and total experience aggregated from previous employments. Dismissal frequencies per company were tracked monthly.

1.3 Summary Statistics

Table 1.2 presents summary statistics that compare the characteristics of the workers in the Austrian workforce in general, all job losers, and the estimation sample specifically used in this replication

study similar to Table.

Table 1.2: Summary Statistics

Variable	All Workers (1994) (1)	All (2)	Estimation Sample (3)
Age in years	32.9	33.2	31.2
Female (%)	48.7	48.3	51.5
Married (%)	–	38.3	43.4
Austrian citizen (%)	–	89.5	88.0
Blue collar occupation (%)	57.3	59.9	57.5
Months of tenure	–	47.0	26.4
Months worked in past five years	–	48.2	42.5
Mean duration of nonemployment (months)	–	15.8	16.9
Observed in new job (%)	–	92.8	92.4
Sample Size	70,058	1,159,903	650,922

Note: Table entries are means unless otherwise noted. Column (1) is based on random sample of all workers between the ages of 20 and 50 in 1994. Column (2) includes individuals losing a job in the private sector over the period 19802001 who are between ages 20 and 50, worked at their previous firm for more than one year, and took up UI benefits within 28 days of job loss (eliminating quitters). Sample in column (3) further eliminates job losers from construction, those who returned to their previous employer, or those who worked for more than five years at their previous firm. Wages expressed in real (year 2000) euros. Nonemployment duration is time from end of lost job to start of next job.

1.4 Validity of Identification

Identification validity relies on continuity assumptions around eligibility thresholds. Potential manipulation by firms around the 36-month tenure threshold could threaten the RD design validity. While explicit clustering of layoffs around this threshold was absent, unobserved firm behaviors and systemic hiring practices, such as preference for temporary or lower bargaining-power workers, could potentially bias outcomes. Rigorous enforcement of Austrian employment regulations and compulsory works council consultations significantly restrict such strategic behaviors. The continuity of covariates and absence of clear layoffs clusters near thresholds support the credibility of the RD approach in estimating causal effects of severance pay and extended unemployment benefits.

Chapter 2

Replication of Main Results

In this chapter, we examine the replication of key findings related to the effects of severance pay and extended unemployment benefits on job-search behavior and reemployment outcomes. Employing a regression discontinuity approach, we analyze shifts in nonemployment durations around specific tenure and work-experience thresholds.

2.1 Graphical Evidence

2.1.1 Layoff Distribution

Figure 2.1 illustrates the frequency of job losers entering unemployment in each 31-day interval around the three-year tenure cutoff. There is no distinct spike in layoffs at the 35-month mark, nor a significant dip just after 36 months, implying that employers do not appear to strategically circumvent severance obligations.

2.1.2 Worker Histories and Wage Patterns

Figure 2.2 depicts the average number of past jobs held by individuals for each tenure-month category. We look for a jump in job histories around 36 months that could suggest selection issues or distinct worker types on either side of the threshold. The figure shows no visible break, indicating that prior job histories do not differ significantly at the cutoff.

Figure 2.3 shows mean wages for those laid off with different months of tenure. There is a slight increase of roughly 300 euros/year in average wages just after 36 months. Although this difference attains statistical significance in a large sample, it represents a small fraction of overall earnings. Thus, while some selection on wages exists, it is not large in economic terms.

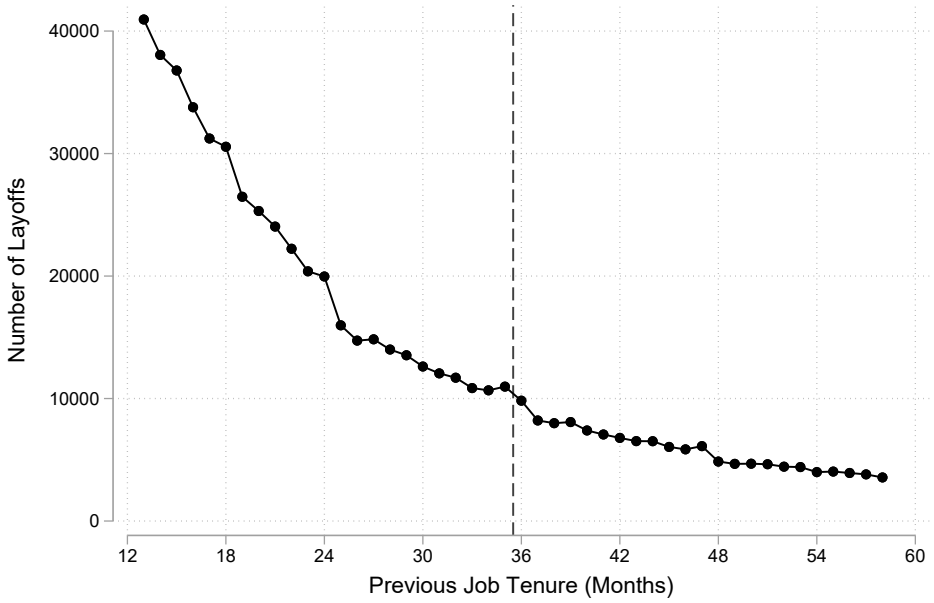


Figure 2.1: Frequency of Layoffs by Job Tenure

Note: Number of Job Losers by Month of Tenure. The months are categorized into 31-day intervals around the three-year threshold. There is no systematic bunching at 35 months of tenure, suggesting that employers do not time layoffs to avoid severance pay.

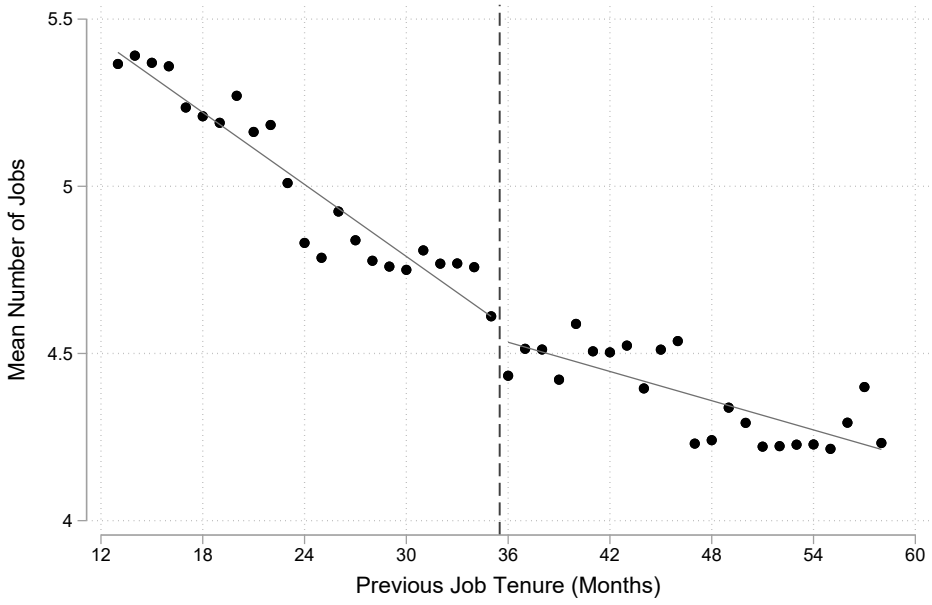


Figure 2.2: Number of Jobs Held by Job Tenure

Note: Observed characteristics such as past job count show no discontinuity at 36 months, suggesting similar job histories for those just above and below the threshold.

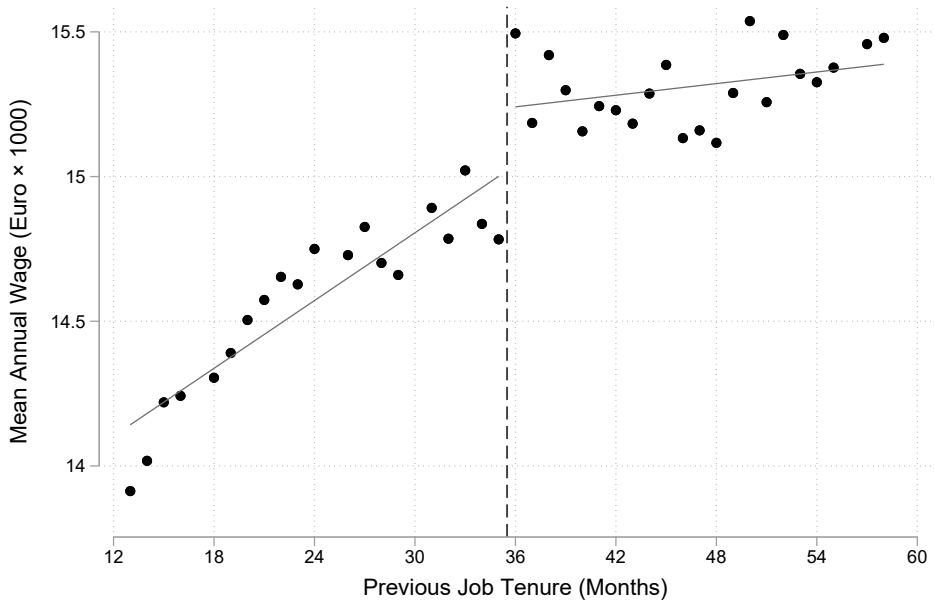


Figure 2.3: Wage by Tenure

Note: There is a small but statistically detectable increase in mean wage after 36 months, equivalent to about 1.6% of average wages.

2.2 Nonemployment Durations and Job-Finding Hazards

2.2.1 Unadjusted Durations

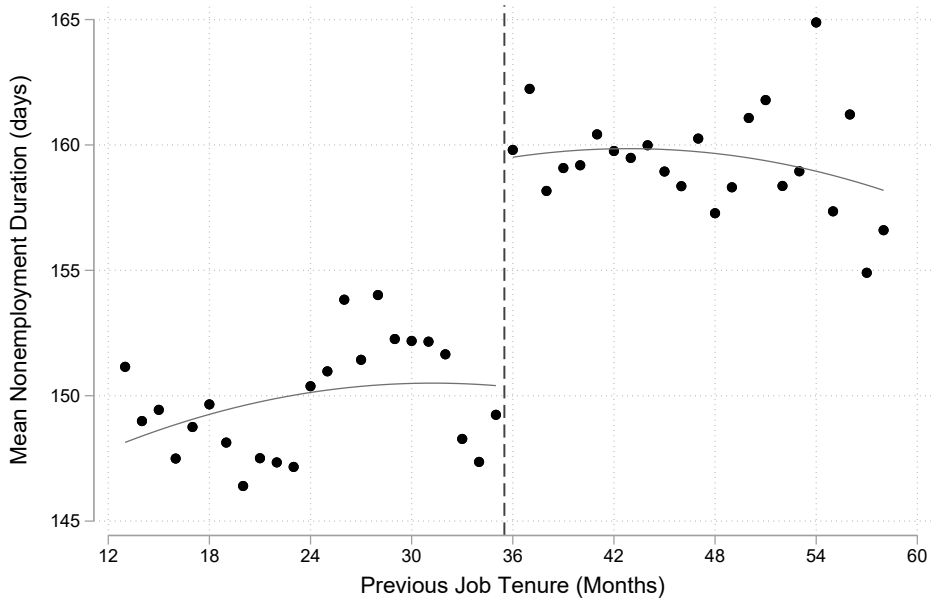


Figure 2.4: Effect of Severance Pay on Nonemployment Durations

Note: Average Nonemployment Durations by Tenure-Month Category (Uncensored). Censoring is not accounted for, and spells exceeding two years are excluded. A polynomial trend is fit separately on each side of the 36-month cutoff.

Figure 2.4 plots the average duration of nonemployment for each tenure-month group without accounting for censoring. Because observations lasting more than two years are excluded, and no adjustments are made for incomplete spells, the figure conflates the effects of severance pay and extended benefits. Nevertheless, it provides an initial sense that job losers with more than three years of tenure display somewhat longer average durations.

2.2.2 Censoring-Adjusted Hazards

Figure 2.5 adjusts for censoring by limiting attention to the first 20 weeks (i.e., 140 days) of unemployment. A Cox model is applied with polynomial controls for tenure and an indicator for extended UI benefits. The drop in hazard at 36 months suggests that those eligible for severance pay exhibit lower job-finding rates in the early weeks of unemployment.

Figure 2.6 presents weekly job-finding hazards for individuals who worked at another firm within the last five years. The two groups (33–35 vs. 36–38 months of tenure) highlight the effect of

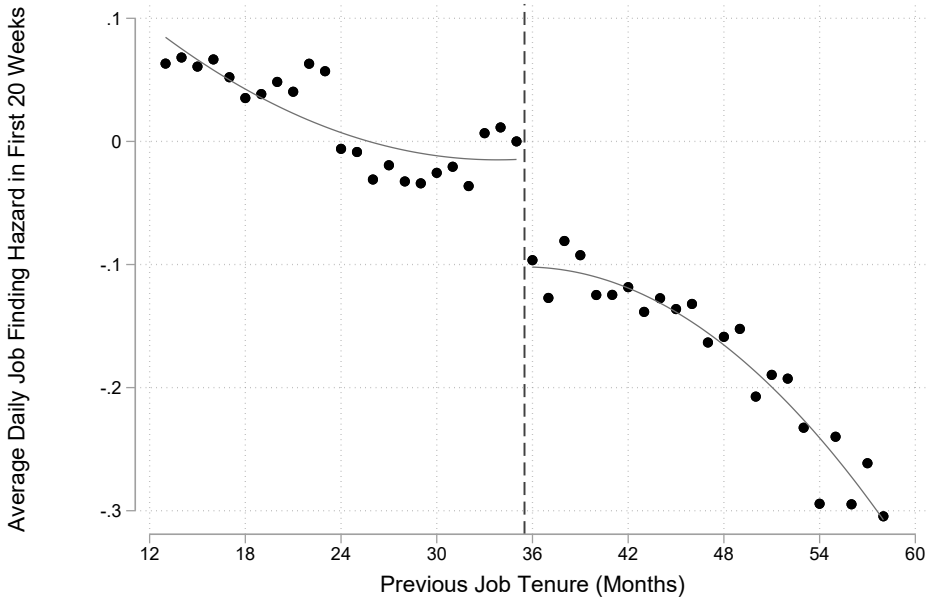


Figure 2.5: Effect of Severance Pay on Job-Finding Hazards

Note: A cubic polynomial in job tenure and interactions with extended benefit eligibility are included, with spells longer than 140 days censored.

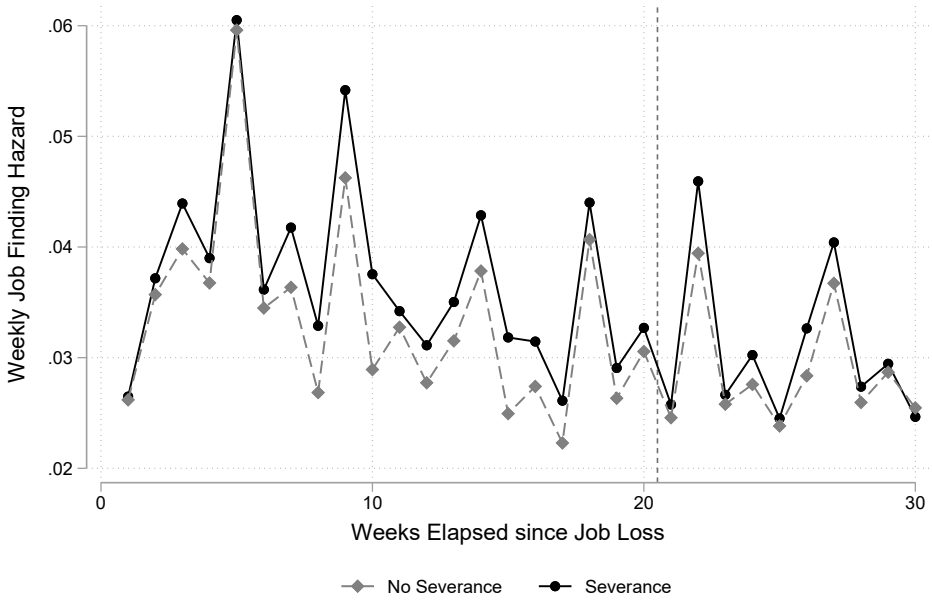


Figure 2.6: Effect of Severance Pay on Job Finding Hazards by Week

Comparing job losers with 33–35 months vs. 36–38 months of tenure shows a persistent gap in reemployment rates.

severance pay eligibility. The hazard gap persists over the first few weeks of the spell, with job losers just above the eligibility threshold exhibiting consistently lower rates of returning to work.

2.2.3 Extended Benefits Effects

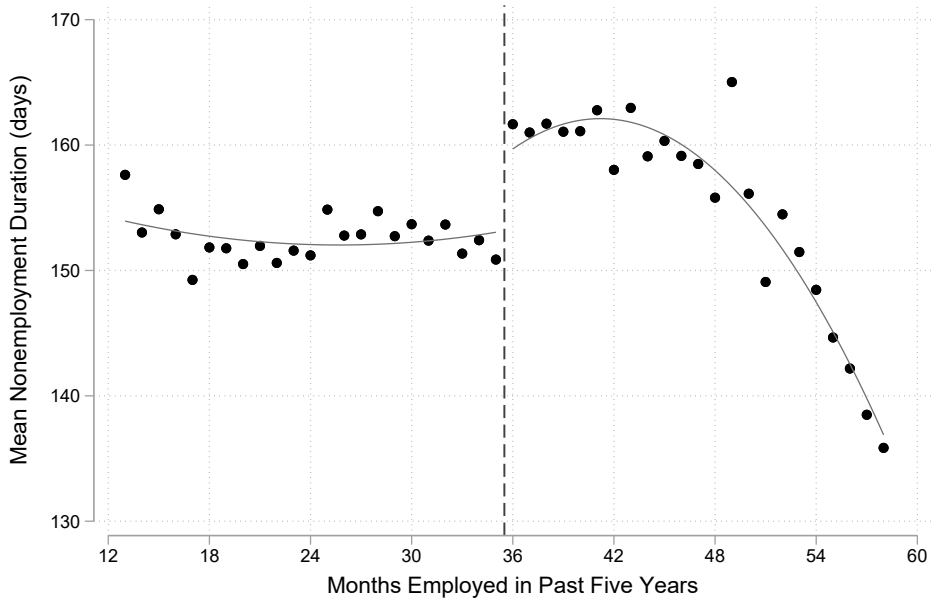


Figure 2.7: Effect of Benefit Extension on Nonemployment Durations

Note: Durations exceeding two years are excluded, and separate polynomial fits are shown around the 36-month cutoff for extended benefits.

Figure 2.7 focuses on individuals grouped by total months of employment in the preceding five years, showing a jump in average nonemployment duration at the 36-month point, where UI benefits increase from 20 to 30 weeks.

In Figure 2.8, the daily hazard rate is again examined for a censored 20-week window, incorporating a polynomial in previous work experience and interactions with severance pay status. The hazard clearly shifts down at 36 months, indicating that individuals with extended UI eligibility search less intensively in the early weeks of unemployment.

Figure 2.9 shows weekly hazards for those who worked at least one month in another firm prior to the last job, splitting them by work experience near the 36-month cutoff. The gap before week 20 is consistent with individuals anticipating their extended benefits, reducing search intensity from the outset.

First, there is no significant gathering around the severance pay eligibility threshold, supporting the notion that firms do not strategically fire employees just before qualifying for severance. Second,

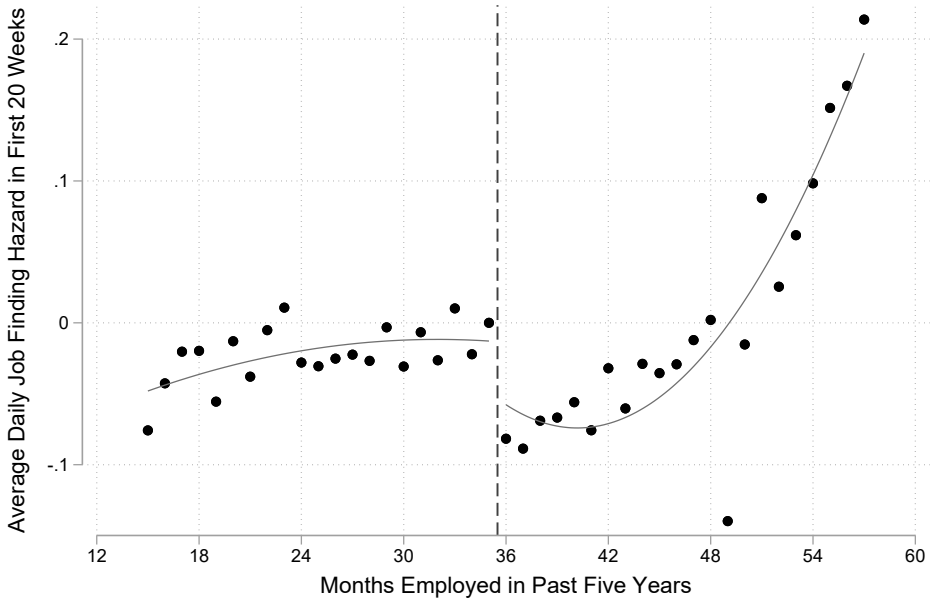


Figure 2.8: Effect of Extended Benefits on Job-Finding Hazards

Note: Daily Hazard Rates for the First 20 Weeks, Controlling for Severance Pay. Estimated with a Cox model and plotted by the months worked in the past five years, revealing a discontinuous shift in reemployment rates at 36 months.

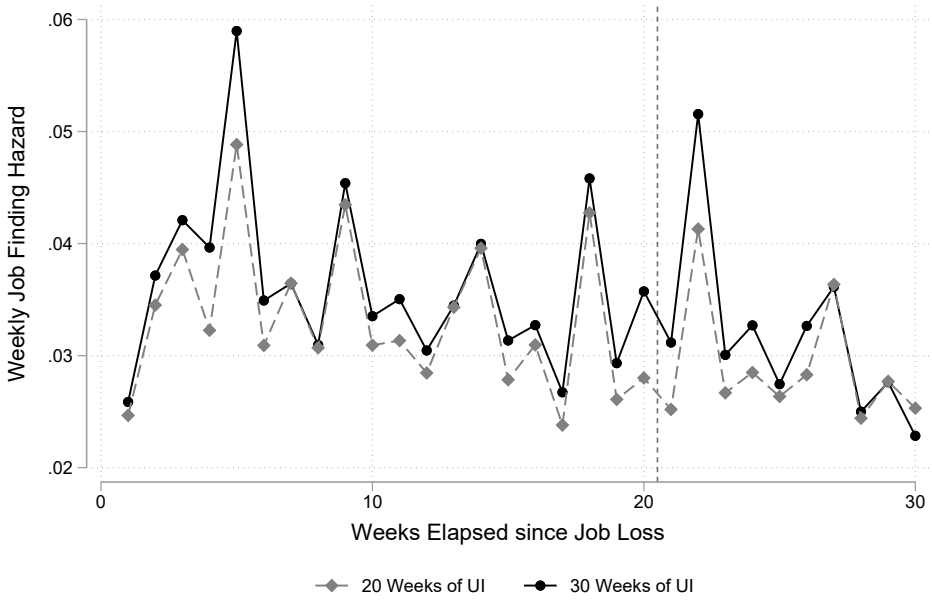


Figure 2.9: Effect of Extended Benefits on Job-Finding Hazards by Week

Note: Weekly Hazard Rates for a Restricted Sample, Separating 20-Week UI from 30-Week UI Groups. Comparing those with 33–35 months of experience vs. 36–38 months highlights the forward-looking behavior of individuals who anticipate extended benefits.

job histories and observable characteristics do not differ sharply around the cutoff, suggesting that selection bias is limited. Third, both severance pay eligibility and extended UI benefits appear to delay reemployment, as shown by lower hazard rates among individuals above the respective thresholds.

From a policy perspective, these findings imply that liquidity effects matter: individuals with a cash-on-hand boost via severance or the prospect of extended benefits reduce their search effort and remain unemployed longer. At the same time, forward-looking behavior is evident, as the extended UI benefits lower job-finding rates even before the extension takes effect. The observed lack of strong job quality gains in other analyses (e.g., wage or subsequent employment stability differences) suggests that these prolonged spells do not necessarily yield better matches.

Chapter 3

Additional Tests

This chapter examines several diagnostic tests for the validity of the regression discontinuity (RD) design, focusing on the distributions of baseline covariates and the robustness of local polynomial estimates under different bandwidth and kernel choices.

3.1 Baseline Covariates and Local Randomization

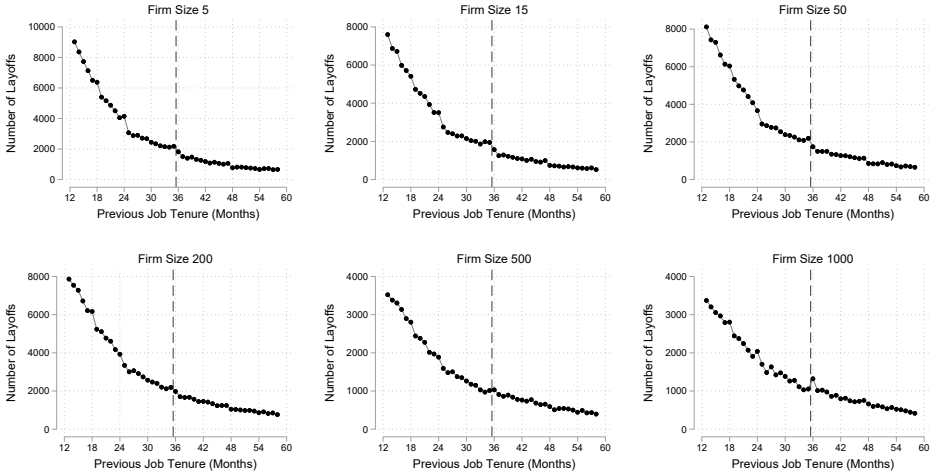
A key assumption for RD identification is that individuals (or firms) cannot perfectly manipulate the running variable to sort selectively into treatment just above (or below) the cutoff point. In this setting, firms might be tempted to dismiss employees just before they reach the 36-month threshold to avoid paying severance. However, if severe manipulation occurred, one would expect a discontinuity in baseline variables (e.g., age, industry, firm size) at the cutoff, indicating non-random assignment near the threshold. Figures 3.1–3.5 plot various baseline characteristics by job tenure:

Figures 3.1–3.5 reveal smooth trends in most baseline variables across the 36-month cutoff, suggesting that employees do not appear systemically different just below or above the threshold. Although small discontinuities can arise (e.g., in age, as shown in Figure 3.4), these effects generally remain small relative to the sample size and do not appear to undermine the continuity assumption. As a result, there is little evidence of significant nonrandom selection close to the cutoff.

3.2 Bandwidth Selection and Local Polynomial Fits

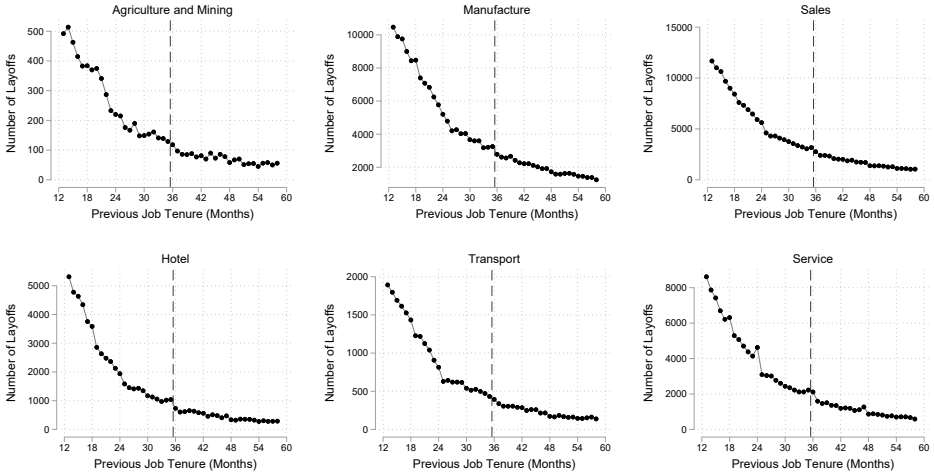
To flexibly estimate the outcome just around the cutoff, we use polynomial approximations in the running variable (previous job tenure) and weight observations based on their distance to the threshold (Lee & Lemieux,). Choosing an appropriate bandwidth for these local polynomial fits is crucial. A large bandwidth uses more data points but may violate the local continuity assumption; a small bandwidth concentrates on observations close to the threshold but can reduce precision.

Figure 3.1: Size of Firms by Job Tenure



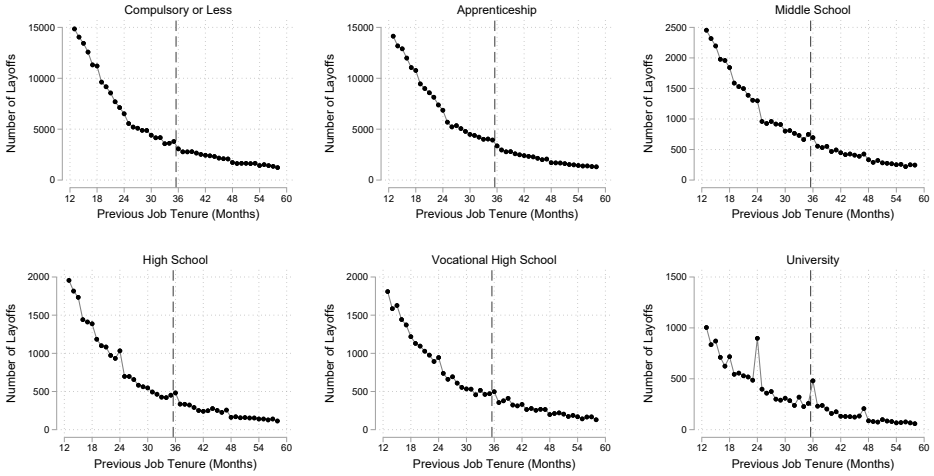
Note: Each panel represents a subset of firms with a given total size (e.g., 5, 15, 50, 200, 500, 1000 employees). The vertical dashed line denotes the 36-month cutoff for severance pay eligibility.

Figure 3.2: Industry by Job Tenure



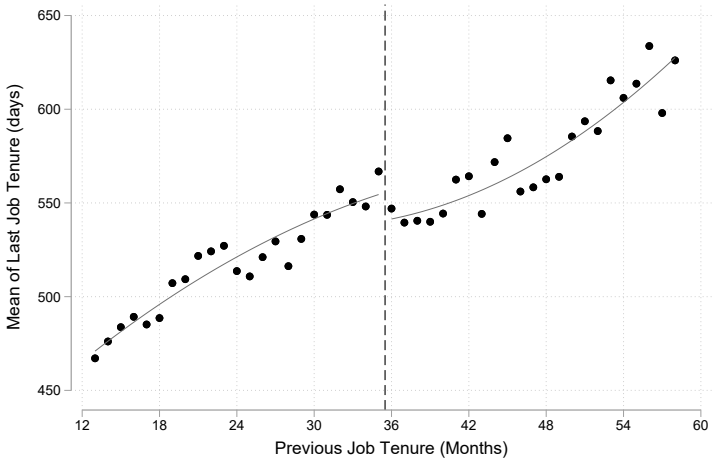
Note: Each panel shows the frequency of layoffs by previous job tenure (in months) for a specific industry. The vertical dashed line is the 36-month threshold for severance eligibility.

Figure 3.3: Education by Job Tenure



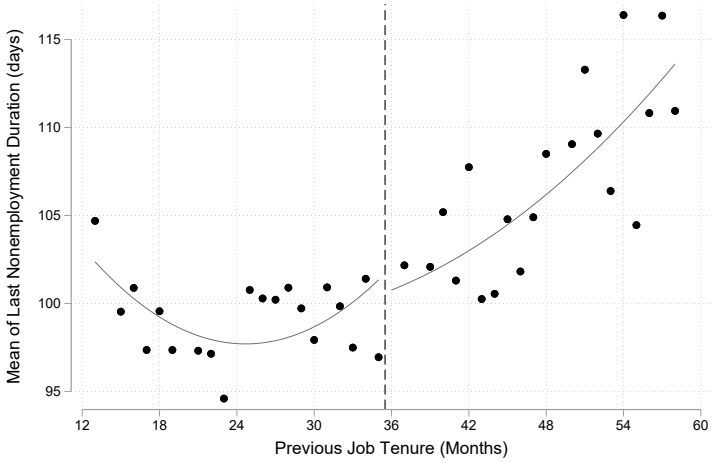
Note: Each panel illustrates the distribution of layoffs over previous job tenure for workers with a particular education level (e.g., compulsory, apprenticeship, secondary, university). The dashed line indicates the 36-month threshold.

Figure 3.4: Age by Job Tenure



Note: This figure plots the mean age against the number of months of previous job tenure. A best-fit line is overlaid separately on each side of the 36-month cutoff.

Figure 3.5: Last Nonemployment Durations by Job Tenure



Note: This figure depicts the average number of days in the last nonemployment spell, by job tenure. The vertical dashed line denotes the 36-month cutoff.

Several kernel weighting functions are common in RD analyses. A *uniform* kernel assigns equal weight to all observations within the bandwidth, whereas a *triangular* kernel gives greater weight to points closer to the cutoff, tapering to zero at the bandwidth boundary. We explore both approaches in this replication.

Optimal Bandwidth Selection

To determine the optimal bandwidth for the regression discontinuity (RD) design, I implemented a loop over different polynomial orders, bandwidth selection methods, and kernel types. This systematic approach balances the bias-variance trade-off, ensuring estimates are both precise and unbiased.

The polynomial order (p) determines the flexibility of the local regression model. The script is designed to iterate over Local linear regression ($p = 1$), Quadratic polynomial ($p = 2$), and Cubic polynomial ($p = 3$). Bandwidth selection is performed using two established methods: MSE-optimal (‘msetwo‘), which minimizes the Mean Squared Error (MSE) while incorporating a bias correction and coverage error rate (‘certwo‘), which selects a bandwidth that minimizes the coverage error rate, ensuring confidence intervals remain valid. Additionally, the kernel function determines how observations near the cutoff are weighted. While the full code includes both uniform and triangular kernels, due to computational constraints, the results presented here are limited to $p = 1, 2$ and uniform kernel. The full version is available in the source code (`do_files/part3_additional_tests.do`) but is commented out to prevent long execution times.

Results of Bandwidth Selection

The table below presents the selected left-side (h_{left}) and right-side (h_{right}) bandwidths for different polynomial orders and selection methods under the uniform kernel:

Table 3.1: Optimal Bandwidth Selection for Local Linear Regression

Polynomial Order	**Method**	h_{left}	h_{right}
1	MSE (msetwo)	23.00	22.00
1	Coverage Error (certwo)	11.80	11.29
2	MSE (msetwo)	23.00	22.00
2	Coverage Error (certwo)	10.72	10.26

These results highlight several important patterns. The optimal bandwidths differ significantly across selection methods. The 'msetwo' method generally results in larger bandwidths ($\sim$23 months on both sides), allowing more observations near the cutoff to be included in estimation. This suggests that incorporating more data, even at the cost of some bias, leads to lower MSE estimates.

For $p = 2$ (quadratic specification), bandwidths remain similar to $p = 1$ under 'msetwo' but are slightly smaller under 'certwo', suggesting that higher-order polynomials may provide more flexibility in capturing functional relationships, leading to slightly more restricted bandwidth choices.

These findings demonstrate the sensitivity of RD estimates to bandwidth choices, reinforcing the need for robustness checks with multiple selection methods and kernel functions. The full set of results, including cubic polynomials ($p = 3$) and triangular kernels, is available in the source code but was excluded from the main results due to computational constraints.

Chapter 4

Hazard Model

Hazard models estimate the likelihood of an event (e.g., employment) occurring over time, given that it has not yet occurred. They are particularly useful for studying unemployment duration data because they can incorporate right-censored observations (those still unemployed at the end of the sample window) in a more flexible way than standard OLS methods. Unlike static regressions, which typically focus on the conditional mean of an outcome, hazard models focus on the conditional hazard rate. This emphasis is especially relevant for questions involving the probability of obtaining a job on day t given having been unemployed up to day $t-1$.

4.1 Cox Proportional Hazard Model

We use a semi-parametric Cox model, which factors the hazard rate into two parts:

$$h(t \mid X_i) \;=\; \lambda_0(t)\; \exp\big(\beta^\top X_i\big),$$

where $\lambda_0(t)$ is the unknown baseline hazard capturing duration dependence, and $\exp(\beta^\top X_i)$ captures the proportional shifts in the hazard attributable to observed covariates X_i. By leaving the baseline hazard $\lambda_0(t)$ unspecified, the Cox model permits flexible forms of duration dependencecritical for analyzing unemployment spells, which can exhibit changing search intensity or employer perceptions over time. In addition, the partial likelihood estimation method in the Cox framework naturally handles right-censored spells without imposing strong parametric assumptions on the shape of the hazard function.

4.2 Estimation Results

Effects on Nonemployment Durations

Table 4.1 reports estimates of severance pay and extended benefit eligibility on nonemployment durations, using a Cox proportional hazards specification. A negative coefficient implies a reduction in the hazard of finding a job (i.e., an increase in expected spell length). Column (1) includes no controls beyond polynomial terms for tenure and months worked, whereas Columns (2) and (3) progressively introduce demographic and job history controls. Column (4) restricts attention to firms that laid off at least four individuals during the sample window, further mitigating concerns about selective firing.

Table 4.1: Effects of Severance Pay and Extended Benefits on Nonemployment Durations: Hazard Model Estimates

	No Controls	Basic Controls	Full Controls	≥ 4 Layoffs by Firm
	(1)	(2)	(3)	(4)
Severance Pay	-0.125***	-0.115***	-0.092***	-0.132**
	(0.017)	(0.018)	(0.019)	(0.058)
Extended Benefits	-0.093***	-0.064***	-0.062***	-0.079
	(0.016)	(0.017)	(0.018)	(0.059)
Sample Size	650,922	565,835	509,355	48,390

Notes: Coefficients are from Cox proportional hazard models censored at 20 weeks. A negative coefficient implies a lower job-finding hazard (i.e., an increase in nonemployment duration). Column (1) includes no controls beyond polynomial terms for tenure/months worked; Columns (2) and (3) add demographic and job history controls. Column (4) further restricts the sample to firms with at least four layoffs. Standard errors (in parentheses) are clustered by individual. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The consistently negative coefficients suggest that both severance pay eligibility and extended UI benefits delay reemployment. For instance, in Column (1), severance pay eligibility reduces the daily job-finding hazard by about 12.5%, while extended benefits reduce it by roughly 9.3%. These estimates remain robust after adding control variables and restricting to larger layoff firms.

Effects on Match Quality

Table 4.2 turns to match-quality outcomes: changes in log wages and the duration of the subsequent job. Columns (1) and (2) focus on wage growth, while Columns (3) and (4) measure the duration of the next job. In all specifications, severance pay and extended benefit eligibility have statistically insignificant, near-zero effects on subsequent wages and job tenure.

This finding indicates that although severance pay and extended benefits prolong initial nonemployment spells, they do not lead to gains (or losses) in subsequent wages or job stability. In other

Table 4.2: Effects of Severance Pay and Extended Benefits on Match Quality

	Dep. Variable: Change in Log Wage		Dep. Variable: Duration of Next Job	
	No Controls (1)	**Full Controls** (2)	**No Controls** (3)	**Full Controls** (4)
Severance Pay	-0.009	-0.005	-0.018	0.004
	(0.007)	(0.006)	(0.014)	(0.015)
Extended Benefits	-0.005	-0.010	-0.005	0.009
	(0.006)	(0.006)	(0.013)	(0.014)
Sample Size	553,607	445,926	601,152	476,307

Notes: Column (1) and (2) use the change in log wage as the outcome; Columns (3) and (4) use the duration of the next job. "No Controls" includes only the treatment indicators and polynomials in tenure/months worked; "Full Controls" adds demographic, job history, industry, and time controls. Standard errors in parentheses, clustered by individual.

words, the extended search time does not appear to translate into better-paying or more durable matches, at least over the horizons measured in this data.

The hazard model evidence implies that both severance pay and extended UI benefits create liquidity effects, lowering the early-job-finding hazard by 6–12%. However, the impact on match quality is minimal, suggesting limited returns to additional search in terms of job stability or post-unemployment wages. Together, these results underscore the importance of taking into account the liquidity effect when analyzing unemployment behavior: while providing a valuable consumption cushion, these benefits may also slow the transition back to work without yielding significant subsequent earnings or job-duration gains.

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